Deliverable 3: Ultrasound Physics & Imaging

Overview

This deliverable covers the complete ultrasound physics and imaging chain, from wave propagation through tissue to image formation. Starting with fundamental acoustic properties and wave equations, we build up to pulse-echo imaging, Doppler ultrasound, and advanced techniques including harmonic imaging and contrast enhancement.

Learning Objectives

- Understand acoustic wave propagation in tissue
- Derive and solve the wave equation for ultrasound
- Calculate acoustic impedance and reflection coefficients
- Design pulse-echo systems with appropriate resolution and penetration
- Implement Doppler ultrasound for blood flow measurement
- Understand harmonic imaging and contrast enhancement
- Apply ultrasound physics to clinical diagnostic applications

Contents Overview

- Part 1: Acoustic Physics & Tissue Interactions
- Part 2: Transducers & Beam Forming
- Part 3: Doppler Ultrasound
- Part 4: Advanced Techniques & Applications

Part 1: Acoustic Physics & Tissue Interactions

Key Concepts

Acoustic impedance, attenuation, and wave propagation in tissue

Learning Objectives

- Understand acoustic wave propagation in tissue
- Derive the wave equation for ultrasound
- Calculate acoustic impedance and reflection coefficients
- Understand attenuation and its frequency dependence
- Apply ultrasound physics to clinical diagnostic applications

Topics Covered

- Acoustic impedance and reflection coefficients
- Attenuation and frequency dependence
- Wave equation for ultrasound
- Tissue properties and acoustic parameters
- Pulse-echo principle
- Resolution and penetration trade-offs

Exercises

Exercise 1: Acoustic Impedance

Given: $\rho_{blood} = 1060$ kg/m³, $c_{blood} = 1570$ m/s, $\rho_{muscle} = 1080$ kg/m³, $c_{muscle} = 1580$ m/s

Find:

$$Z_{blood} = \rho_{blood} c_{blood} = 1060 \times 1570 = 1.664 \times 10^6 \text{ Rayls}$$

$$Z_{muscle} = \rho_{muscle} c_{muscle} = 1080 \times 1580 = 1.706 \times 10^6 \text{ Rayls}$$

$$R = \frac{Z_2 - Z_1}{Z_2 + Z_1} = \frac{1.706 \times 10^6 - 1.664 \times 10^6}{1.706 \times 10^6 + 1.664 \times 10^6} = \frac{0.042 \times 10^6}{3.370 \times 10^6} = 0.0125$$

$$T = \frac{2Z_2}{Z_2 + Z_1} = \frac{2 \times 1.706 \times 10^6}{3.370 \times 10^6} = 1.013$$

$$\frac{I_r}{I_i} = R^2 = 0.0125^2 = 0.0156 \text{ (1.56\%)}$$

$$\frac{I_t}{I_i} = \frac{4Z_1 Z_2}{(Z_2 + Z_1)^2} = \frac{4 \times 1.664 \times 10^6 \times 1.706 \times 10^6}{(3.370 \times 10^6)^2} = \frac{11.37 \times 10^{12}}{11.36 \times 10^{12}} = 1.000 \text{ (100\%)}$$

- Acoustic impedance mismatch causes reflection at tissue boundaries
- Even small impedance differences produce measurable echoes
- Transmission remains high despite reflection

Exercise 2: Attenuation

Given: $\alpha = 0.5$ dB/cm/MHz, $f = 5$ MHz, $d = 10$ cm

Find:

$$\alpha_{total} = \alpha \cdot f = 0.5 \text{ dB/cm/MHz} \times 5 \text{ MHz} = 2.5 \text{ dB/cm}$$

$$\text{Attenuation} = \alpha_{total} \cdot d = 2.5 \text{ dB/cm} \times 10 \text{ cm} = 25 \text{ dB}$$

$$\frac{I_t}{I_i} = 10^{-\text{Attenuation}/10} = 10^{-25/10} = 10^{-2.5} = 0.0032 \text{ (0.32\%)}$$

- Attenuation increases with frequency and depth
- Only 0.32% of ultrasound reaches 10 cm depth at 5 MHz
- Higher frequencies provide better resolution but limited penetration

Implementation

- Python implementation: exercise_1.py

Visualization

Clinical Relevance

- Understanding tissue boundaries for diagnostic interpretation
- Optimizing frequency for depth vs resolution trade-off
- Contrast agent design for enhancing echoes

Exercise 3: Tissue Transmission Model

Given: Layer 1: $\rho_1 = 1000$ kg/m³, $c_1 = 1500$ m/s, $d_1 = 2$ cm; Layer 2: $\rho_2 = 1100$ kg/m³, $c_2 = 1600$ m/s, $d_2 = 3$ cm; $f = 5$ MHz

Find:

$$Z_1 = \rho_1 c_1 = 1000 \times 1500 = 1.500 \times 10^6 \text{ Rayls}$$

$$Z_2 = \rho_2 c_2 = 1100 \times 1600 = 1.760 \times 10^6 \text{ Rayls}$$

$$R_{12} = \frac{Z_2 - Z_1}{Z_2 + Z_1} = \frac{1.760 \times 10^6 - 1.500 \times 10^6}{1.760 \times 10^6 + 1.500 \times 10^6} = \frac{0.260 \times 10^6}{3.260 \times 10^6} = 0.0798$$

$$T_{12} = \frac{2Z_2}{Z_2 + Z_1} = \frac{2 \times 1.760 \times 10^6}{3.260 \times 10^6} = 1.080$$

$$\alpha_{total} = \alpha \cdot f \cdot d = (0.5 \times 5 \times 2) + (0.6 \times 5 \times 3) = 5 + 9 = 14 \text{ dB}$$

$$\frac{I_t}{I_i} = 10^{-14/10} \cdot (1 - R_{12}^2) = 0.0398 \times (1 - 0.0798^2) = 0.0398 \times 0.9936 = 0.0395 \text{ (3.95\%)}$$

- Multi-layer tissue model with different acoustic properties
- Cumulative attenuation through all layers
- Total transmission accounts for both attenuation and reflection

Exercise 4: Resolution

Given: $f = 5$ MHz, $c = 1540$ m/s, $\lambda = 0.308$ mm

Find:

$$\text{Axial resolution} = \frac{\lambda}{2} = \frac{0.308 \text{ mm}}{2} = 0.154 \text{ mm}$$

$$\text{Lateral resolution} \approx \lambda \cdot F\#$$

- Axial resolution depends only on wavelength
- Lateral resolution depends on focusing (F#)
- Higher frequency improves both but reduces penetration

Implementation

- Python implementation: exercise_3.py

Visualization

Clinical Relevance

- Multi-layer tissue models for realistic simulation
- Frequency selection for different clinical applications
- Understanding why deep tissues require lower frequencies

Part 2: Transducers & Beam Forming

Key Concepts

Piezoelectric transducers, beam forming, and focusing

Learning Objectives

- Understand piezoelectric transducer operation
- Calculate beam focusing parameters
- Design phased array systems
- Understand apodization and sidelobe control

Topics Covered

- Piezoelectric transducer operation
- Beam focusing and steering
- Phased array systems
- Apodization and sidelobe control
- Resolution vs penetration trade-off

Exercises

Exercise 1: Focusing

Given: $f = 5$ MHz, $D = 20$ mm, $c = 1540$ m/s

Find:

$$\lambda = \frac{c}{f} = \frac{1540 \text{ m/s}}{5 \text{ MHz}} = 0.308 \text{ mm}$$

$$F\# = \frac{F}{D} = \frac{50 \text{ mm}}{20 \text{ mm}} = 2.5$$

$$\text{Focal spot width} \approx \lambda \cdot F\# = 0.308 \text{ mm} \times 2.5 = 0.77 \text{ mm}$$

$$\text{Depth of field} \approx 4\lambda \cdot F\#^2 = 4 \times 0.308 \text{ mm} \times 2.5^2 = 7.7 \text{ mm}$$

- Focusing improves lateral resolution
- Depth of field limits the focused region
- Trade-off between focal depth and resolution

Implementation

- Python implementation: exercise_2.py

Visualization

Clinical Relevance

- Transducer selection for different applications
- Focal zone placement for optimal resolution
- Understanding why deep tissues require lower frequencies

Part 3: Doppler Ultrasound

Key Concepts

Doppler effect, blood flow measurement, and color flow imaging

Learning Objectives

- Understand Doppler effect in ultrasound
- Calculate velocity from Doppler shift
- Design pulse-Doppler systems
- Understand aliasing and its limits
- Implement color flow imaging

Topics Covered

- Doppler effect in ultrasound
- Velocity measurement
- Pulse-Doppler systems
- Aliasing and Nyquist limit
- Color flow imaging
- Power Doppler
- Tissue Doppler imaging

Exercises

Exercise 1: Doppler Shift

Given: $f_0 = 5$ MHz, $v = 0.5$ m/s, $\theta = 60°$, $c = 1540$ m/s

Find:

$$f_D = \frac{2 f_0 v \cos\theta}{c} = \frac{2 \times 5 \text{ MHz} \times 0.5 \text{ m/s} \times \cos 60°}{1540 \text{ m/s}} = \frac{5 \times 10^6 \times 0.5 \times 0.5}{1540} = 1.62 \text{ kHz}$$

$$v = \frac{f_D c}{2 f_0 \cos\theta} = \frac{1.62 \text{ kHz} \times 1540 \text{ m/s}}{2 \times 5 \text{ MHz} \times \cos 60°} = 0.5 \text{ m/s}$$

- Doppler shift proportional to velocity
- Angle dependence requires correction
- Positive shift for approaching, negative for receding

Implementation

- Python implementation: exercise_1.py

Visualization

Clinical Relevance

- Blood flow velocity measurement
- Cardiac function assessment
- Vascular disease detection
- Fetal monitoring

Part 4: Advanced Techniques & Applications

Key Concepts

Harmonic imaging, contrast agents, and advanced techniques

Learning Objectives

- Understand harmonic generation in tissue
- Design contrast agent detection
- Implement harmonic imaging
- Understand elastography principles

Topics Covered

- Harmonic imaging
- Contrast agents
- Elastography
- 3D ultrasound
- Clinical applications

Exercises

Exercise 1: Harmonic Generation

Given: $f_0 = 2$ MHz, $A_1 = 1$, $A_2 = 0.1$, $A_3 = 0.01$

Find:

$$\text{Fundamental} = A_1 = 1$$

$$\text{2nd harmonic} = \frac{A_2}{A_1} = \frac{0.1}{1} = 10\%$$

$$\text{3rd harmonic} = \frac{A_3}{A_1} = \frac{0.01}{1} = 1\%$$

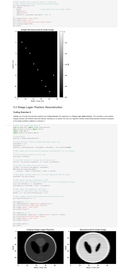
$$\text{THD} = \sqrt{A_2^2 + A_3^2} = \sqrt{0.1^2 + 0.01^2} = 0.1005 = 10.05\%$$

- Harmonics generated by tissue nonlinearity
- Second harmonic typically strongest
- Harmonic imaging improves contrast and resolution

Implementation

- Python implementation: exercise_2.py

Visualization



Clinical Relevance

- Harmonic imaging improves contrast and resolution
- Contrast agents enhance blood flow visualization
- Elastography assesses tissue stiffness
- 3D ultrasound provides volumetric assessment

Part 5: Doppler Imaging & Artifact Analysis

Key Concepts

Aliasing, clutter, and artifacts in Doppler ultrasound

Learning Objectives

- Understand aliasing and Nyquist limit in Doppler
- Design clutter rejection filters
- Recognize and avoid common artifacts
- Optimize Doppler parameters for clinical applications

Topics Covered

- Aliasing and Nyquist limit
- Clutter rejection filters
- Artifacts and their recognition
- Optimization strategies
- Clinical pitfalls and limitations

Exercises

Exercise 1: Aliasing

Given: $f_0 = 5$ MHz, $c = 1540$ m/s, $v_{max} = 1$ m/s, $\theta = 0°$

Find:

$$f_{max} = \frac{2 f_0 v_{max} \cos\theta}{c} = \frac{2 \times 5 \text{ MHz} \times 1 \text{ m/s} \times \cos 0°}{1540 \text{ m/s}} = 6.49 \text{ kHz}$$

$$\text{PRF} \geq 2 f_{max} = 2 \times 6.49 \text{ kHz} = 13 \text{ kHz}$$

- Aliasing occurs when velocity exceeds Nyquist limit
- Increasing angle reduces maximum detectable velocity
- Higher PRF prevents aliasing but limits depth

Implementation

- Python implementation: exercise_1.py

Visualization

Clinical Relevance

- Aliasing in high velocity flow
- Clutter from tissue motion
- Artifacts affecting diagnosis
- Optimization for clinical applications

Part 6: Contrast Imaging & Advanced Techniques

Key Concepts

Contrast agents, harmonic imaging, and molecular imaging

Learning Objectives

- Understand contrast agent physics
- Design harmonic imaging systems
- Understand molecular imaging principles
- Apply contrast imaging to clinical applications

Topics Covered

- Contrast agent physics
- Harmonic imaging
- Molecular imaging
- Perfusion imaging
- Clinical applications

Exercises

Exercise 1: Contrast Enhancement

Given: $I_{baseline} = 100$, $I_{contrast} = 400$, $\Delta t = 30$ s

Find:

$$\text{Enhancement} = \frac{I_{contrast} - I_{baseline}}{I_{baseline}} = \frac{400 - 100}{100} = 300\%$$

$$\text{Enhancement ratio} = \frac{I_{contrast}}{I_{baseline}} = \frac{400}{100} = 4$$

- Contrast agents enhance echo intensity
- Enhancement depends on concentration and time
- Perfusion imaging tracks tissue enhancement

Implementation

- Python implementation: exercise_1.py

Visualization

Clinical Relevance

- Contrast enhancement improves diagnostic accuracy
- Harmonic imaging reduces tissue clutter
- Molecular imaging enables targeted diagnosis
- Perfusion imaging assesses tissue viability

Part 7: Clinical Applications & Case Studies

Key Concepts

Real-world applications, diagnostic protocols, and clinical decision making

Learning Objectives

- Understand clinical ultrasound applications
- Design diagnostic protocols
- Interpret ultrasound images
- Make clinical decisions based on ultrasound findings

Topics Covered

- Abdominal imaging
- Cardiac imaging
- Vascular imaging
- Fetal imaging
- Tissue characterization
- Quality assessment

Exercises

Exercise 1: Abdominal Imaging

Given: Liver lesion, $f = 3.5$ MHz, $d = 15$ cm

Find:

$$\text{Attenuation} = 0.5 \times 3.5 \times 15 = 26.25 \text{ dB}$$

$$\frac{I_t}{I_i} = 10^{-26.25/10} = 0.0024 \text{ (0.24\%)}$$

- Deep lesions require lower frequency
- Attenuation limits penetration
- Trade-off between resolution and penetration

Implementation

- Python implementation: exercise_1.py

Visualization

Clinical Relevance

- Abdominal imaging for liver, kidney, gallbladder, pancreas
- Cardiac imaging for echocardiography, valve disease, cardiomyopathy, congenital heart
- Vascular imaging for carotid artery, deep vein thrombosis, peripheral vascular
- Fetal imaging for fetal anatomy, fetal heart, placenta
- Tissue characterization for liver fibrosis, tumor characterization, elastography
- Quality assessment for image quality, diagnostic accuracy, protocol compliance

Summary

Key Takeaways

- Acoustic impedance mismatch causes reflection at tissue boundaries
- Attenuation increases with frequency and depth
- Transducer focusing improves lateral resolution
- Doppler shift proportional to velocity
- Harmonic imaging improves contrast and resolution
- Contrast agents enhance blood flow visualization
- Clinical applications require understanding of limitations

Next Steps

- Complete all exercises
- Review clinical applications
- Prepare for Deliverable 4: Ultrasound Physics & Imaging

Resources

- Python implementation: exercise_1.py, exercise_2.py, exercise_3.py
- Visualizations: 01-acoustic-properties.jpg, 02-transducer-beam.jpg, 03-doppler-shift.jpg, 04-advanced-techniques.jpg
- Clinical applications: 05-clinical-applications.jpg
- Artifacts and limitations: 06-artifacts-limitations.jpg
- Contrast imaging: 07-contrast-imaging.jpg
- Case studies: 08-case-studies.jpg

Summary Generated

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